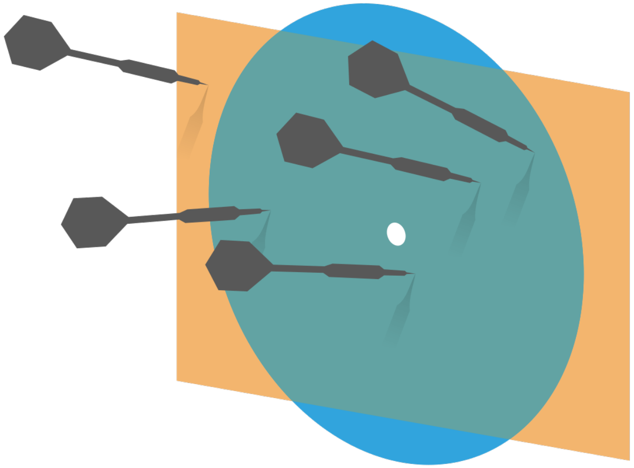
Probability and Statistics for Computer Science



Credit: wikipedia

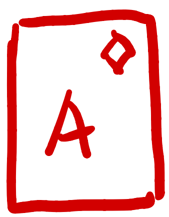
"A major use of probability in statistical inference is the updating of probabilities when certain events are observed" – Prof. M.H. DeGroot

Hongye Liu, Teaching Assistant Prof, CS361, UIUC, 09.12.2024

Event, Joint event

① What does it mean to say an event has happened?

② Joint event | intersection of events

* 

* {HT}

Last time

- More Probability Calculation
- Conditional Probability
  - Multiplication rule
  - Bayes rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Objectives

- Conditional Probability
  - Review
    - Definition
    - Bayes rule
  - Total probability
  - Independence

Which is Larger?

Given $P(B) > 0$ $\quad P(B) < 1$

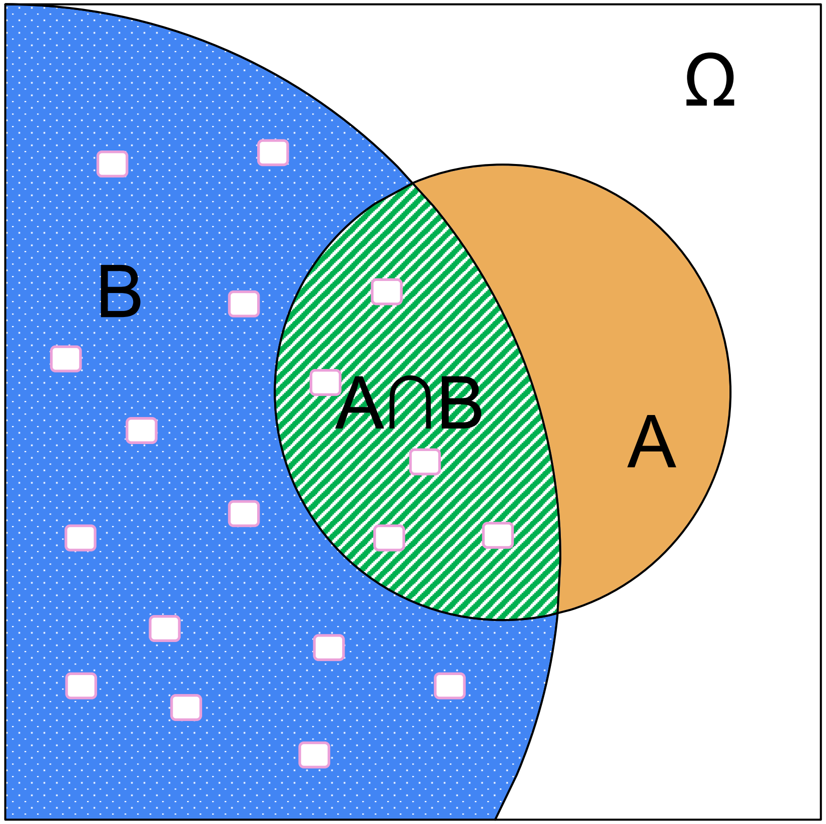
① $P(A \cap B)$

$\boxed{② \; P(A \mid B)}$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability

- The probability of ***A*** given ***B***



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 0$$

The area defined by the set B is the new sample space for the conditional probability P(A| B)

Joint Probability Calculation (chain rule)

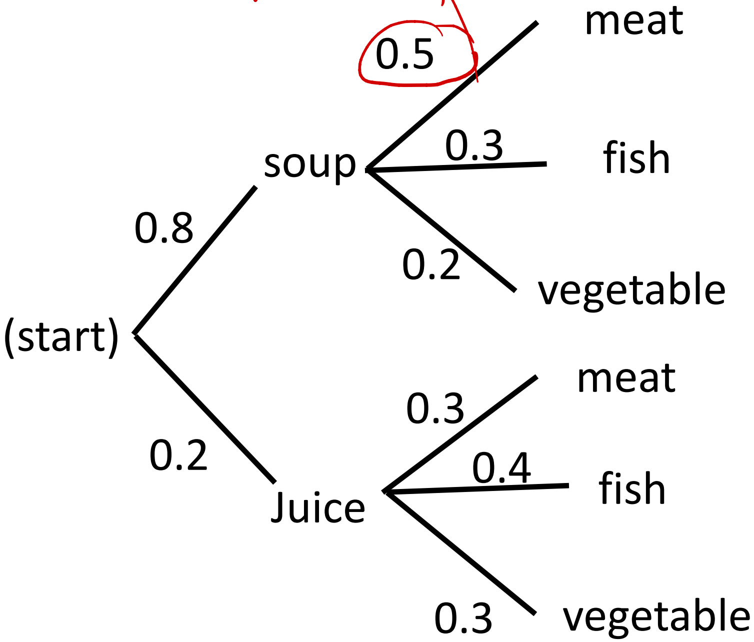
$$\Rightarrow P(A \cap B) = P(A|B)P(B)$$

$p(E_1 \cap E_2 \cap E_3)$

$= p(E_3 | E_1 \cap E_2) p(E_1 \cap E_2)$

$= p(E_3 | E_1 \cap E_2) \cdot p(E_2 | E_1) p(E_1)$

$p(\text{meat} | \text{soup})$

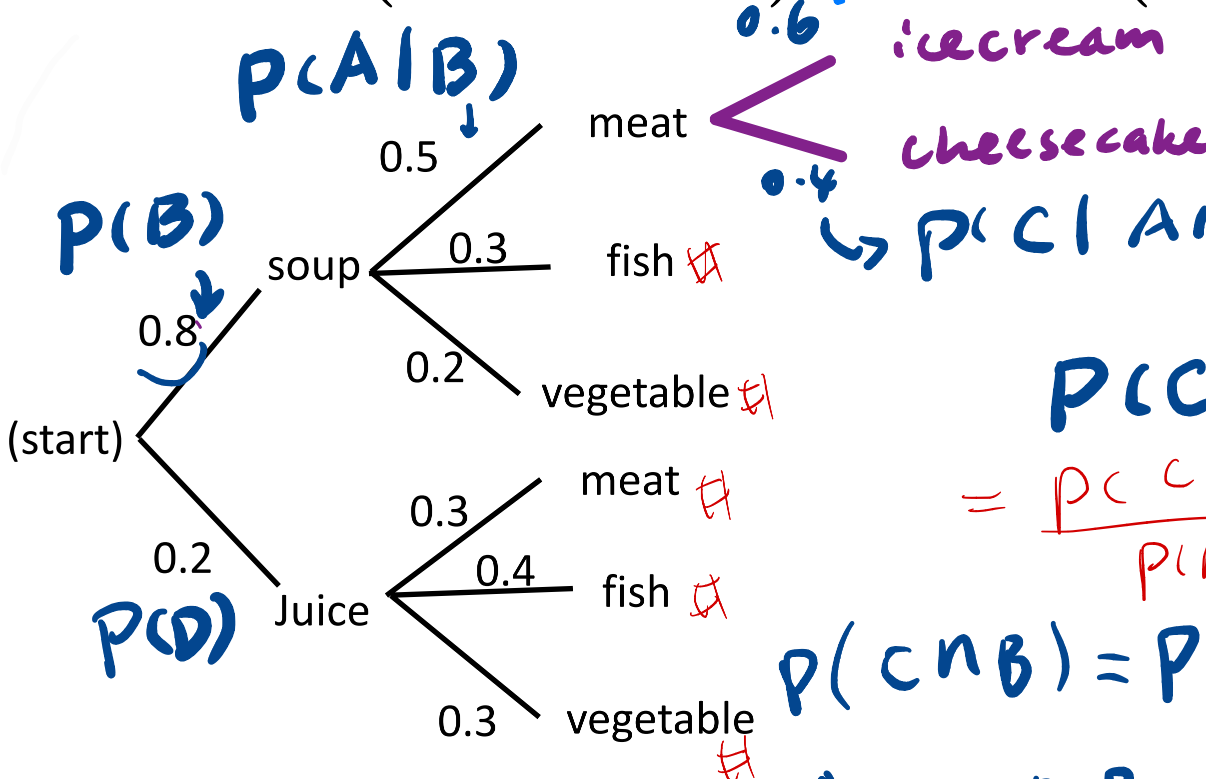


$$P(soup \cap meat) =$$
$$P(meat|soup)P(soup)$$
$$= 0.5 \times 0.8 = 0.4$$

Multiplication using conditional probability

$P(\bar{C}|A\cap B)$

$$P(A\cap B) = P(A|B)P(B)$$



$P(C|A\cap B)$ C: cheesecake

$P(C|B) = ?$

$= \frac{P(C\cap B)}{P(B)} = \frac{0.16}{0.8}$ 2 0

$P(C\cap B) = P(B\cap A\cap C \cup \cancel{B\cap A\cap C})$

A: meat B: soup D: juice

Symmetry of joint event in terms of conditional prob.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

$$\Rightarrow P(A \cap B) = P(A|B)P(B)$$
$$\Rightarrow P(B \cap A) = P(B|A)P(A)$$

Symmetry of joint event in terms of conditional prob.

$B \cap A = A \cap B$

$$\because P(B \cap A) = P(A \cap B)$$

$$\Rightarrow$$

$$\boxed{P(A|B)P(B) = P(B|A)P(A)}$$

The famous **Bayes** rule

$$P(A|B)P(B) = P(B|A)P(A)$$

$$\Rightarrow \quad P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Thomas Bayes (1701-1761)

Bayes rule: lemon cars

There are two car factories, **A** and **B**, that supply the same dealer. Factory **A** produced **1000** cars, of which **10** were lemons. Factory **B** produced **2** cars and both were lemons. You bought a car that turned out to be a lemon. What is the probability that it came from factory **B**?

E_1 : Lemon car E_2 : From B

$$P(E_2|E_1) = \frac{P(E_1|E_2)P(E_2)}{P(E_1)} = \frac{1 \times \frac{2}{1002}}{\frac{12}{1002}} = \frac{1}{6}$$

Total probability:

Sunny : 80% to workout

Rainy : 30% to workout

Snowy : 50% to workout

#Sunny / #Rainy / #Snowy = 5 : 3 : 2

$P(\text{work out}) = ?$

$E_1 \cup E_2 \cup E_3$

$$80\% \times \frac{5}{10} + 30\% \times \frac{3}{10} + 50\% \times \frac{2}{10}$$

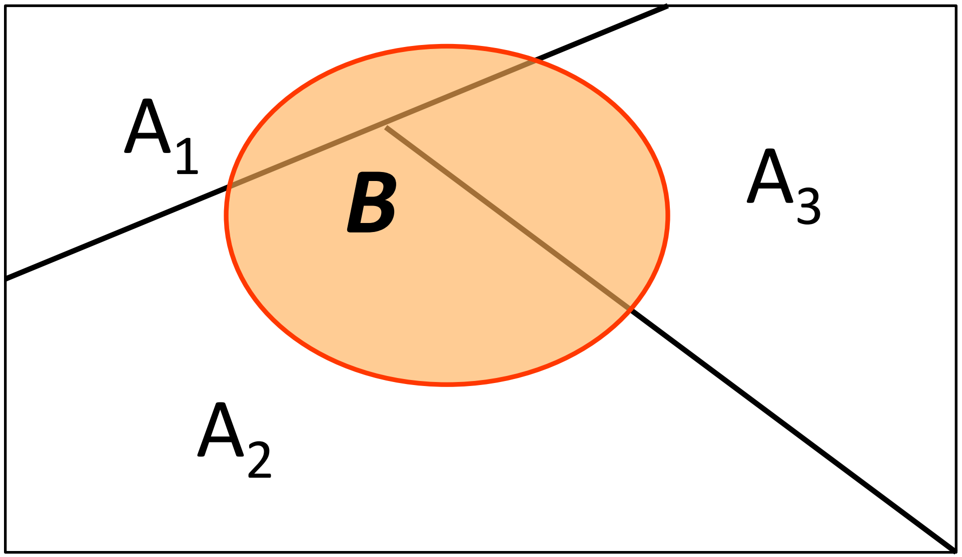
$= 0.59$

(work out ~~on~~ $\cap$ Sunny D)

$= P(\text{Work out} \mid \text{Sunny}) \cdot P(\text{Sunny})$

Total probability

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$
$$= P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)$$



A_1, A_2, A_3 are disjoint,

$A_1 \cup A_2 \cup A_3 \supset B$

Total probability:

$$P(B) = P(A \cap B) + P(A^c \cap B)$$
$$= P(B|A)P(A) + P(B|A^c) \cdot P(A^c)$$

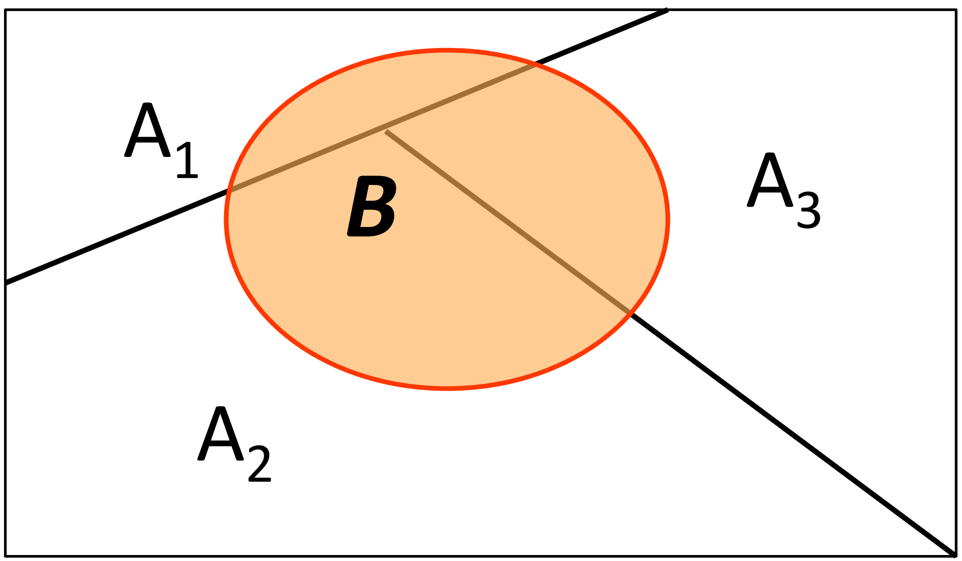
$P(B|A)$ is given

$P(B|A^c)$ is given

Total probability general form

$$P(B) = \sum_j P(B \cap A_j)$$

$$= \sum_j P(B | A_j) P(A_j)$$

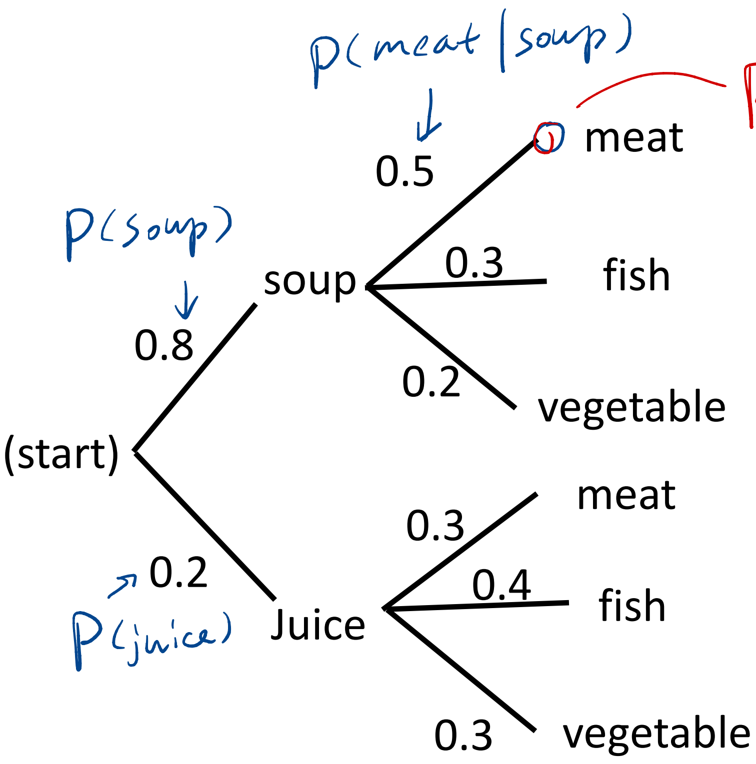


A_j, A_k are disjoint

$A_j \cap A_k = \phi$

$j \neq k$

Total probability:



$P(\text{meat}) = ?$

0.8×0.5 Soup $\cap$ meat

$+ 0.2 \times 0.3$ Juice $\cap$ meat

$= 0.46$

$P(\text{Soup} \cap \text{meat})$

$= P(\text{Meat} \mid \text{Soup}) \times P(\text{Soup})$

$P(\text{soup} \mid \text{meat}) = ?$

$$= \frac{P(\text{soup} \cap \text{meat})}{P(\text{meat})}$$

Bayes rule using total prob.

$$P(A_j | B) = \frac{P(B | A_j)\, P(A_j)}{P(B)}$$

$$= \frac{P(B | A_j)\, P(A_j)}{\sum_j P(B | A_j)\, P(A_j)}$$

$A_i \cap A_j = \phi$ → disjoint

if $i \neq j$

Bayes rule: rare disease test

There is a blood test for a rare disease. The frequency of the disease is 1/100,000. If one has it, the test confirms it with probability 0.95. If one doesn't have, the test gives false positive with probability 0.001. What is $P(D|T)$, the probability of having disease given a positive test result?

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

Using total prob.

$95\% \times 1 \times 10^{-5}$

$+ (\quad)(1-P(D))$

0.001

$$= \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)}$$

Bayes rule: rare disease test

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$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)}$$

$$= \frac{0.95 \times 10^{-5}}{0.95 \times 10^{-5} + 0.001 \times (1 - 10^{-5})} < 1\%$$

Calculate Conditional Prob.

① $P(A|B)$ Counting! i.e. Insurance prob.

② $P(A|B) = \dfrac{P(A\cap B)}{P(B)}$

③ $P(A|B) = \dfrac{P(B|A)P(A)}{P(B)}$

④ $P(A|B) = \dfrac{P(B|A)P(A)}{\sum_j P(B|A_j)P(A_j)}$

A_i, A_j disjoint

$\bigcup_i A_j \supseteq B$

Independence

- One definition:

$$P(A|B) = P(A) \; or$$
$$P(B|A) = P(B)$$

$P(A) > 0$ & $P(B) > 0$

Whether A happened doesn't change the probability of B and vice versa

Independence: example

- Suppose that we have a fair coin and it is tossed twice. let A be the event "the first toss is a head" and B the event "the two outcomes are the same."

$P(A|B)$ $P(A)$

$\underline{P(B|A)} = \frac{P(B \cap A)}{P(A)}$ $P(B) = 50\%$

$= 0.5$ $= \frac{P(\{HH\})}{0.5} = \frac{1/4}{0.5} = 0.5$

- These two events are independent!

Independence

- Alternative definition

$$P(A|B) = P(A)$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = P(A)$$

$$\Rightarrow \boxed{P(A \cap B) = P(A)P(B)}$$

Testing Independence:

- Suppose you draw one card from a standard deck of cards. E_1 is the event that the card is a King, Queen or Jack. E_2 is the event the card is a Heart. Are E_1 and E_2 independent?

$P(E_1 \cap E_2)$

$\frac{3}{52}$

$P(E_1) \times P(E_2)$

$\frac{12}{52} \times \frac{1}{4}$

$= \frac{3}{52}$

indpt !!

Pairwise independence is not mutual independence in larger context

A_1 ♥	A_2 ♦
A_4 ♣	A_3 ♠

P(A_1) = P(A_2) = P(A_3) = P(A_4) = 1/4

$P(A) = ?$ 0.5 $P(B) = ?$ 0.5 $P(C) = ?$ 0.5

$$A = A_1 \cup A_2; P(A)$$

$$B = A_1 \cup A_3; P(B)$$

$$C = A_1 \cup A_4; P(C)$$

$P(A \cap B) = ?$ $\frac{1}{4}$ $P(A)P(B) = ?$ $\frac{1}{4}$

$P(A \cap C) = ?$ $\frac{1}{4}$ $P(A)P(C) = ?$ $\frac{1}{4}$

$P(B \cap C) = ?$ $\frac{1}{4}$ $P(B)P(C) = ?$ $\frac{1}{4}$

$P(A \cap B \cap C) = ?$ $\frac{1}{4}$ $P(A)P(B)P(C) = ?$ $\frac{1}{8}$

Mutual independence

- Mutual independence of a collection of events $A_1, A_2, A_3 \ldots A_n$ is :

$$P(A_i | A_j \cap A_k \cap \ldots A_p) = P(A_i)$$

$$j, k, \ldots p \neq i$$

- It's very strong independence!

Probability using the property of Independence: Airline overbooking (1)

- An airline has a flight with 6 seats. They always sell 7 tickets for this flight. If ticket holders show up independently with probability **p**, what is the probability that the flight is overbooked ?

Mutual

$P(\text{overbooked}) = p^7$

$\overbrace{p \cdot p \cdot p \cdots p}^{n=7}$

Probability using the property of Independence: Airline overbooking (2)

- An airline has a flight with 6 seats. They always sell 8 tickets for this flight. If ticket holders show up independently with probability **p**, what is the probability that exactly 6 people showed up?

P(6 people showed up) = $\binom{8}{6} p^6 \cdot (1-p)^2$

Conditional Independence

- Event ***A*** and ***B*** are conditional independent given event ***C*** if the following is true.

$$P(A \cap B|C) = P(A|C)P(B|C)$$

See an example in Degroot et al. Example 2.2.10

Assignments

- Module week3 on Canvas
- Next time: Random variable

Additional References

- Charles M. Grinstead and J. Laurie Snell "Introduction to Probability”
- Morris H. Degroot and Mark J. Schervish "Probability and Statistics”

See you next time

See You!